Solutions

4.18 Probability Review (Algebra)

1. Events A and B are independent. It is given that $\mathrm{P}(A) = 2\,\mathrm{P}(B)$ and $\mathrm{P}(A \cup B) = 0.52$. Let $x = \mathrm{P}(B)$.

   (a) Show that $2x^2 - 3x + 0.52 = 0$. [3]

   (b) Hence find $\mathrm{P}(B)$. [2]

   (c) Find $\mathrm{P}(A \cap B)$. [1]

a) $P(A) = 2P(B) = 2x$

$P(A \cap B) = P(A)P(B) = 2x \cdot x$

$= 2x^2$

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0{,}52$

$= 2x + x - 2x^2 = 0{,}52$

$2x^2 - 3x + 0{,}52 = 0$

b) $x = \dfrac{-(-3) \pm \sqrt{(-3)^2 - 4(2)(0{,}52)}}{2(2)}$

$= \dfrac{3 \pm 2.2}{4}$ disregard $x > 1$

$P(B) = 0.2$

(c) $P(A \cap B) = 2(0{,}2)(0{,}2)$

$= 0.08$

2. In a school, 100 students were surveyed about activities A and B.

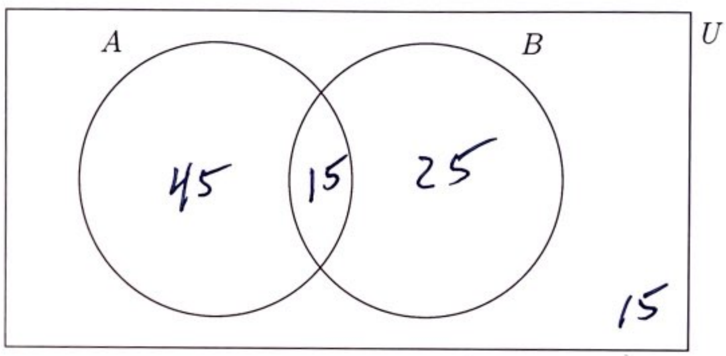
- $\mathrm{P}(A) = 0.6$
- $\mathrm{P}(B \mid A) = 0.25$
- $\mathrm{P}(A' \cap B') = 0.15$

(a) Find $\mathrm{P}(A \cap B)$. [2]

(b) Find $\mathrm{P}(B)$. [2]

(c) Complete the Venn diagram below, showing the number of students in each region. [2]

(d) Determine whether events A and B are independent. Justify your answer. [2]



a) $P(B|A) = \dfrac{P(A\cap B)}{P(A)}$

$P(A\cap B) = P(A)\cdot P(B|A) = 0.6\cdot 0.25$

$= 0.15$

b) ~~A(~~ $P(A\cup B) = 0.6 + P(B) - 0.15 = 1 - 0.15$

$P(B) = 0.4$

c) $P(A\cap B) = 0.15$

$P(A)\,P(B) = 0.6 \cdot 0.4 = 0.24$

$0.15 \neq 0.24$ NOT INDEPENDENT